



QACT2754

**Human Computer Interaction
Individual Assignment**

Coursework Title : Human Computer Interaction

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This form must be completed and submitted alongside any assignment, project, or research work where AI tools have been used. The purpose of this disclosure is to ensure transparency and uphold academic integrity. Failure to disclose AI use may be treated as an academic integrity violation.

Section 1: Student / Researcher Information

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Course/Programme: UKDTPS

Title of Assignment/Research Work: Human Computer Interaction Project Proposal – Trivia Arena

Section 2: Declaration of AI Use

Please tick all that apply and provide details of the AI tools used:

- ☐ No AI tools were used.
- ☐ AI tools were used for supportive tasks (e.g., grammar correction, reference management, formatting).
- ☒ AI tools were used for information search or summarisation (with verification of accuracy).
- ☐ AI tools were used for data analysis support (e.g., code generation, statistical syntax assistance).
- ☒ AI tools were used for content creation support (e.g., brainstorming ideas, structuring outlines).
- ☒ AI tools were used for image, audio, or video generation (with clear attribution).

Section 3: Details of AI Tools Used

Please specify the tool(s) used (e.g., ChatGPT, Grammarly, DALL-E, etc.), the purpose of use, and how the final submission reflects your own original work.

Tool(s) Used: ChatGPT, Claude

Purpose of Use:

- (i) image, audio, or video generation
- (ii) information search or summarization
- (iii) content creation support

Section 4: Student / Researcher Declaration

I hereby declare that:

1. The final submission is my own original work and reflects my own understanding and thought process.
2. All AI use has been disclosed honestly and accurately.
3. I take full responsibility for the accuracy, validity, and integrity of the work submitted.

Signature: Nicholas Paul Goh Chang Yew

Date: 12/8/2026

Trivia Arena

A Turn-Based Quiz Battle Experience with AI-Generated Characters

Human-Computer Interaction — Submitted for Lecturer Approval

11-Week Solo Project | Plain HTML, CSS & JavaScript

1. Executive Summary

Trivia Arena is an interactive, browser-based HCI project that merges the mechanics of turn-based combat games with the educational value of knowledge quizzes. Two players create personalized pixel art characters generated from their own selfies via AI, choose 4 combat moves each, and battle each other in a Street Fighter-inspired arena. The core innovation is a question-gated attack system: every move must be earned by answering a general knowledge question under time pressure.

The game introduces three interlocking engagement mechanics — a 25-second answer timer, a steal phase for wrong answers, and a streak system that rewards consecutive correct answers with a critical hit. A 3-minute battle clock adds urgency and ends in a dramatic sudden-death HP drain if no winner is decided in time.

The entire project is built using plain HTML, CSS, and JavaScript. No frameworks, no server, no installation required. Hence, anybody with stable network connection and access to any web browser will be able to play this game. The project is completed over 11 weeks at one hour of work per day (~77 hours total).

Sustainable Development Goals:

SDG 4: Quality Education

By transforming passive knowledge recall into an active, consequence-driven interaction. Unlike traditional quiz tools where wrong answers carry no stakes, this game's steal mechanic and streak system create intrinsic motivation to engage with each question seriously."

SDG 9: Industry, Innovation and Infrastructure

By building a novel interaction model that combines gaming and education — that's technological innovation. Your use of AI-generated pixel art sprites is also a creative application of emerging AI tools.

SDG 10: Reduced Inequalities

By using free tools throughout — Open Trivia DB, plain HTML/CSS/JS, free Vercel hosting. Anyone with a browser can play it. No paywalls, no downloads, no expensive hardware needed.

2. Problem Statement

Two well-established categories of interactive software suffer from opposite engagement problems:

- Educational quiz tools are passive. A student answers a question, sees a result, and moves on. There are no stakes, no opponent, and no tension. Without meaningful consequences, correct answers feel routine and wrong answers feel inconsequential. Engagement drops quickly and completion rates remain low.
- Competitive games are highly engaging but educationally empty. Games like Pokemon and Street Fighter reward reflex and strategy but teach nothing transferable beyond the game world.

This project addresses both problems simultaneously. By attaching a quiz mechanic to something players genuinely care about — defeating an opponent in battle — every question suddenly matters. A wrong answer does not just lose a point; it hands the opponent a steal opportunity. A streak of correct answers does not just feel satisfying; it charges a critical hit. The battle context makes the knowledge meaningful.

3. Proposed Solution

3.1 Game Overview

The game runs in a web browser and supports two players on the same device, passing it between turns. A complete session has four phases:

1. **Character Creation** — Each player uploads a selfie. An AI model (Replicate API) generates a pixel art sprite of them as a game character in Street Fighter style. The player enters their name and preferred gender of their character. Character data is saved to localStorage.
2. **Category Selection** — Each player picks exactly 1 category from a list.
3. **Move Selection** — Each player picks exactly 4 combat moves from a catalogue, organised by type: assault, arcane and guardian.
4. **Battle** — Players take turns attacking. Every attack is gated by a timed knowledge question. The outcome of the question determines what happens to the attack. A 3-minute background timer adds urgency throughout.
5. **Resolution** — The player whose HP hits 0 first loses. If the 3-minute timer expires, sudden death drains both HP bars simultaneously. Equal HP at drain end results in a draw.

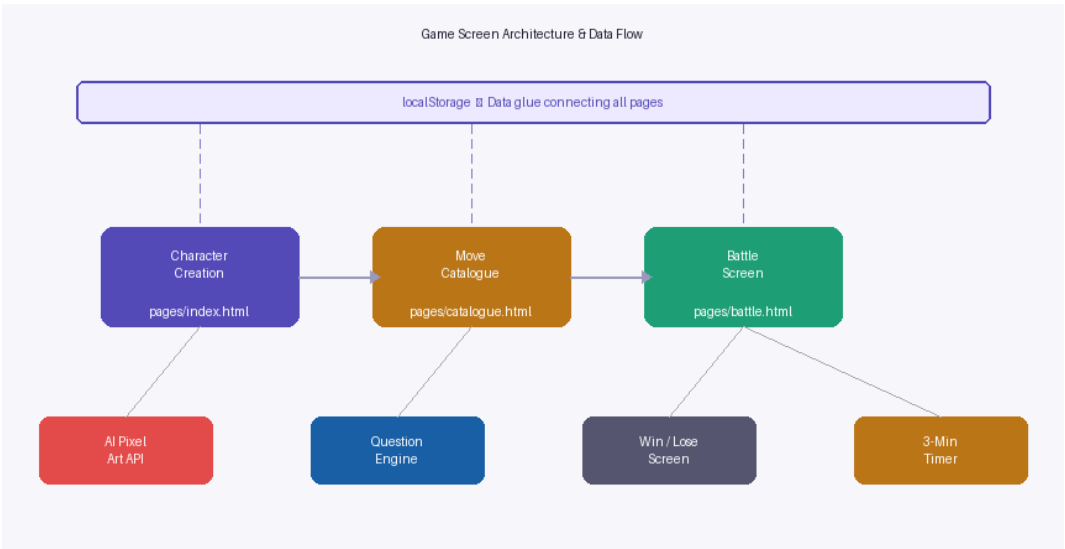


Figure 1 — Game screen architecture and localStorage data flow

3.2 The Question-Gated Attack System

The question mechanic is the primary HCI contribution of this project. Before every attack land, the active player sees a multiple-choice question with a 25-second countdown. There are two possible outcomes:

CORRECT	The attack deals full damage. The active player's streak counter increments by 1. At 3 consecutive correct answers, the next attack automatically deals 1.5× critical damage and the streak resets.
WRONG / TIMEOUT	No damage is dealt. The active player's streak resets to 0. A steal phase immediately activates — the opponent gets 5 seconds to answer the same question. Correct steal = +10% bonus stored for their next attack. Both fail = no damage, turn passes.

The binary design (correct vs wrong) was chosen deliberately over a partial-damage model. Two clean outcomes are easier to understand at a glance, produce clearer emotional feedback, and are simpler to implement and balance correctly.



Figure 2 — Question mechanic: correct answer, wrong answer, steal phase, and streak paths

3.3 The Streak System

Every correct answer in a row increments a visible counter above the active player's character. At 3 consecutive correct answers, a CRIT READY badge appears and the next attack deals $1.5\times$ damage. Design rules:

- Streak increments only on the active player's own correct answer.
- Streak resets to 0 only when the active player answers wrong or times out.
- A failed steal by the opponent does NOT reset the active player's streak.
- The streak counter is visible to both players at all times, creating mounting psychological pressure as it approaches 3.

3.4 The 3-Minute Battle Timer & Sudden Death

A silent countdown runs from 3:00 throughout the entire battle, turning red and pulsing when 30 seconds remain. When it reaches zero:

- "TIME'S UP!" — a full-screen popup appears for 2 seconds.
- Both HP bars begin draining simultaneously at 5 HP per 200ms.
- The player whose HP hits 0 first loses. If both hit 0 on the same tick, the result is a draw.
- If a question is mid-answer when time expires, the current question resolves first — sudden death triggers immediately after the turn completes.

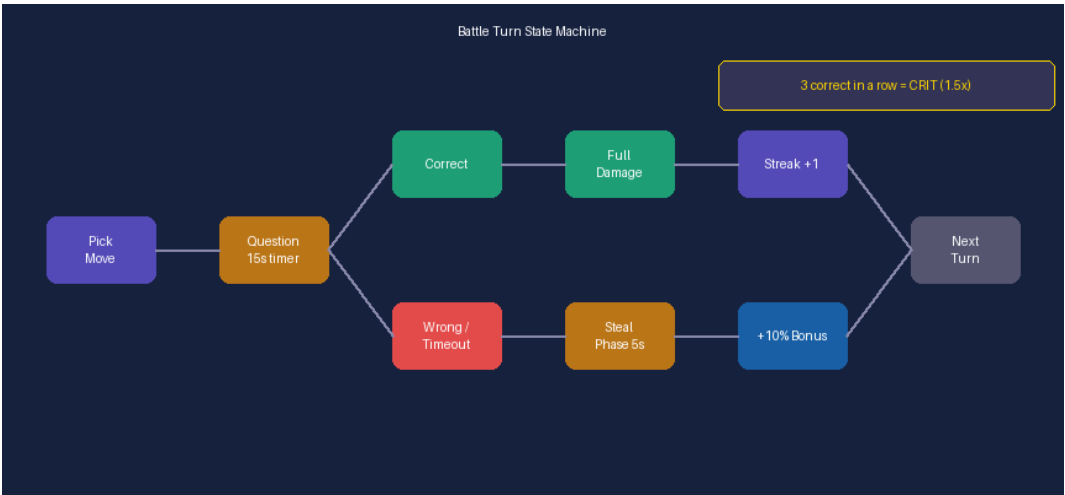


Figure 3 — Full battle turn state machine: move selection through to next turn

3.5 Battle Screen Wireframe



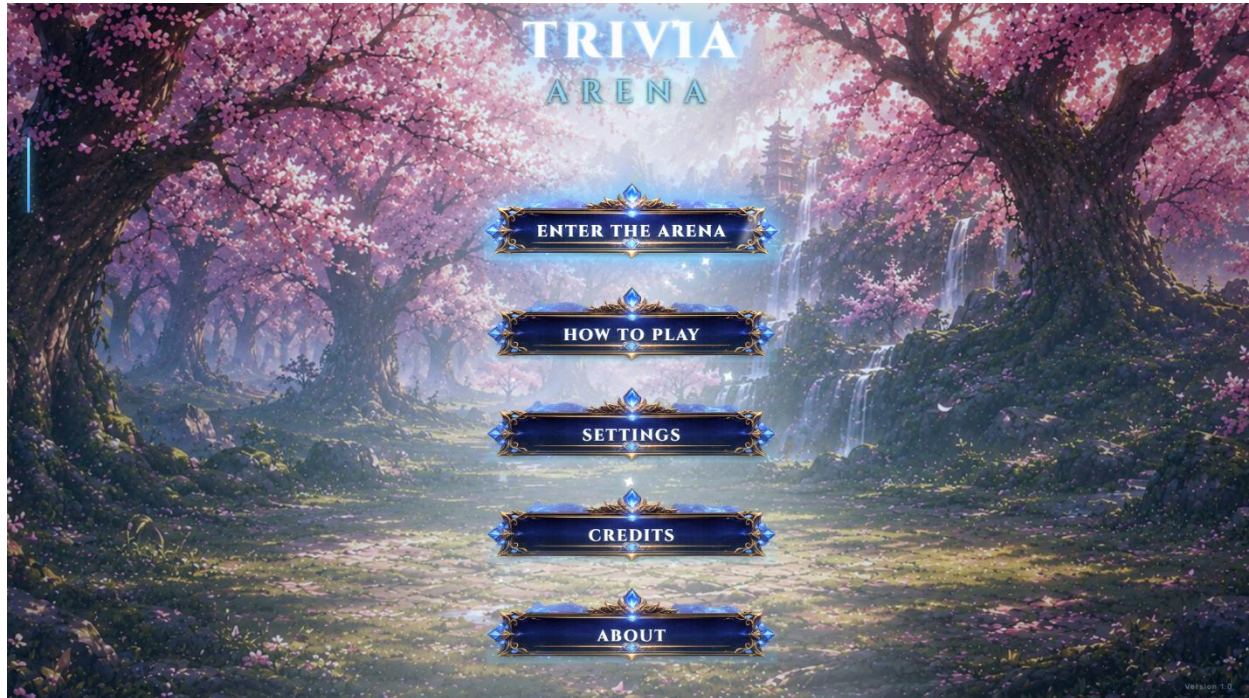
Figure 4 — Battle screen wireframe: fighter zones, HP bars, streak counters, move grid, and question overlay

4. HCI Design Principles & Justifications

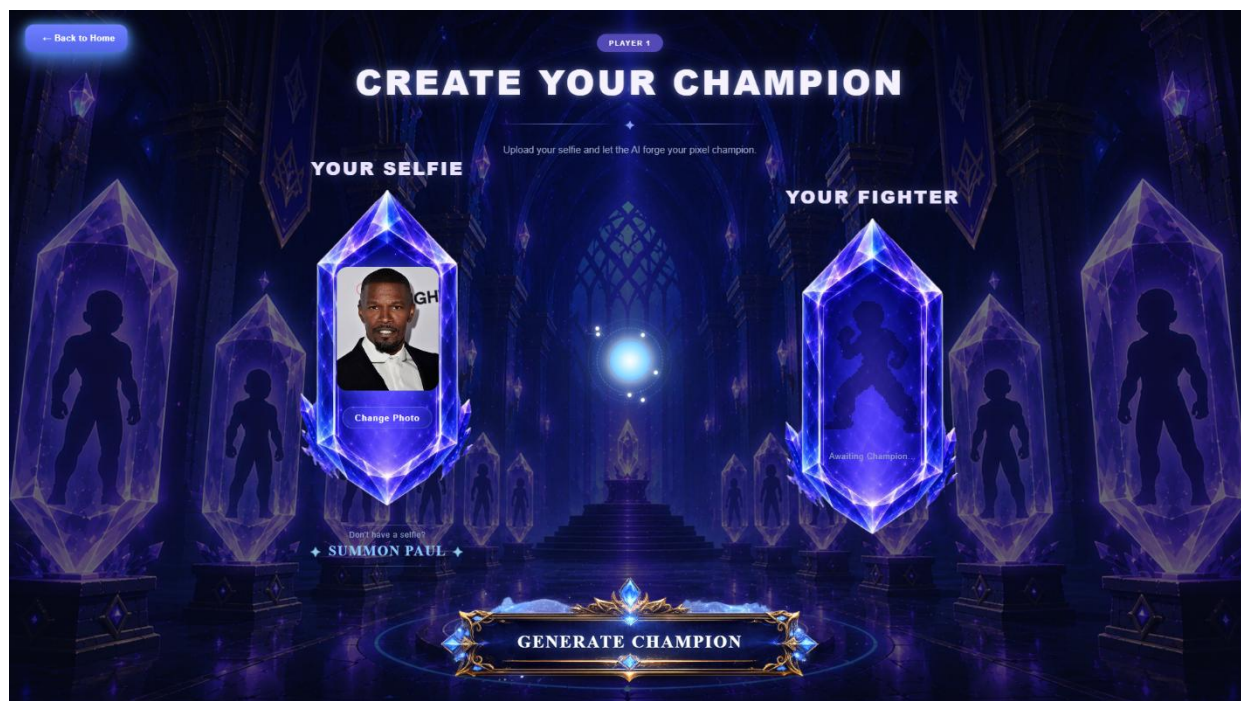
Every mechanic maps to an established HCI principle. The table below documents the design rationale for each decision:

Mechanic	HCI Principle	Justification
Question timer (15s)	Feedback & Urgency	Immediate consequence for inaction. The countdown creates time pressure that elevates even easy questions into tense moments (Norman: feedback must be immediate and informative).
Steal phase	Engagement / Flow	The inactive player is never merely watching. Giving them a stake in every question sustains Flow state (Csikszentmihalyi) for both players simultaneously.
Streak counter	Variable Ratio Reinforcement	A counter visible to both players creates mounting tension. The reward (crit) is predictable in structure but uncertain in timing — the strongest engagement pattern in behavioural learning theory.
Binary outcome (correct/wrong)	Learnability & Clarity	Two clean outcomes are understood within one turn. Partial-damage systems require explanation. Simplicity reduces cognitive load (Miller's Law: limit choices to what users can process instantly).
+10% steal bonus	Perceived Fairness	Small enough not to decide the game, large enough to feel meaningful. Gives the losing player a route back without breaking competitive balance.
AI-generated sprite	Personalisation & Identity	Players see themselves in the game. Named custom characters increase emotional investment and ownership of outcomes (Self-Determination Theory: autonomy and identity).
3-minute battle timer	Pacing & Tension Design	Prevents stalling and guarantees every battle has a climax. Sudden death creates a shared dramatic peak that both players experience simultaneously.
HP colour shift (green→red)	Progressive Disclosure	Critical information (low HP) is revealed at the moment it becomes relevant, not all at once. Reduces cognitive load while increasing situational awareness.

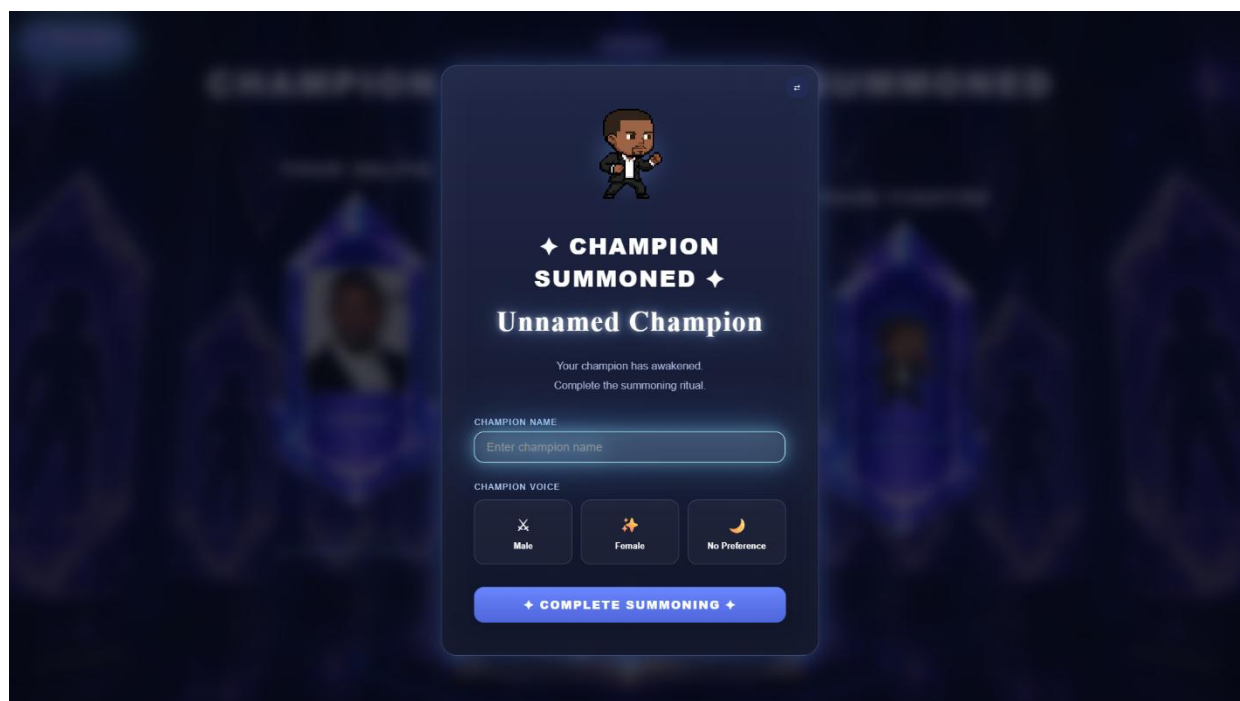
5. Storyboard



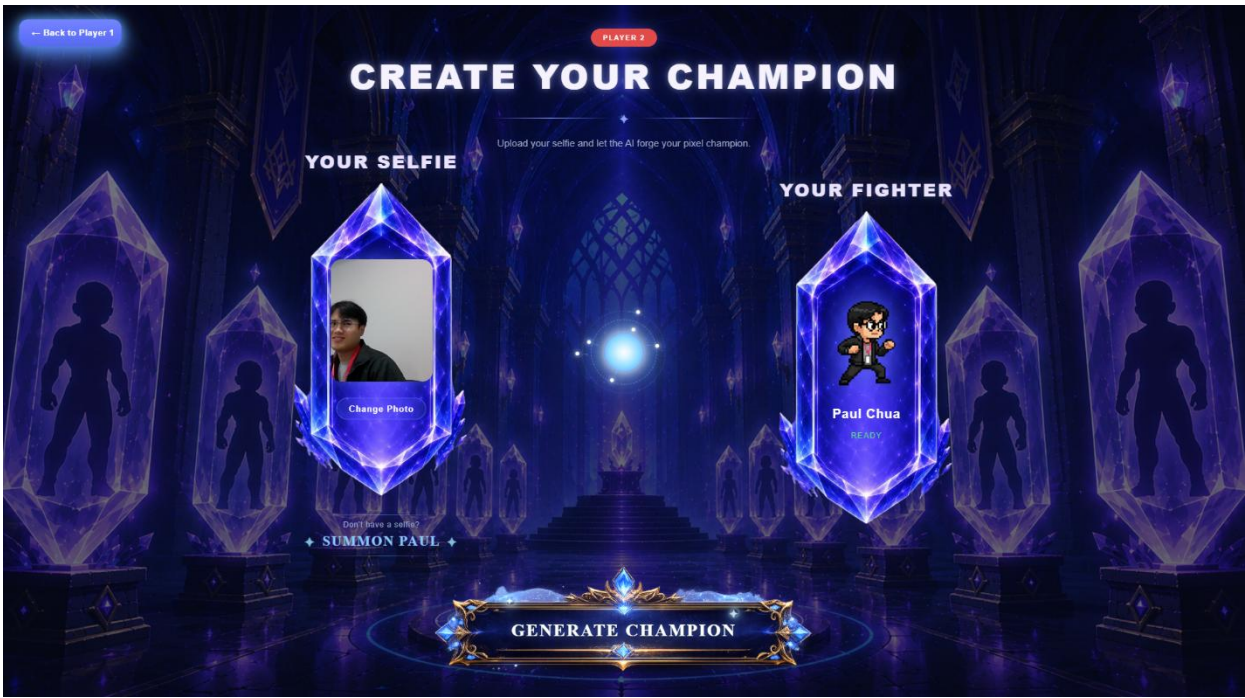
The fantasy-themed home screen welcomes players with glowing crystal buttons, animated scenery, and magical atmosphere, offering quick access to battles, tutorials, settings, credits, and project information.



Player 1 uploads a selfie while an animated crystal chamber displays the future fighter, combining artificial intelligence, fantasy aesthetics, and interactive customization before entering the arena.

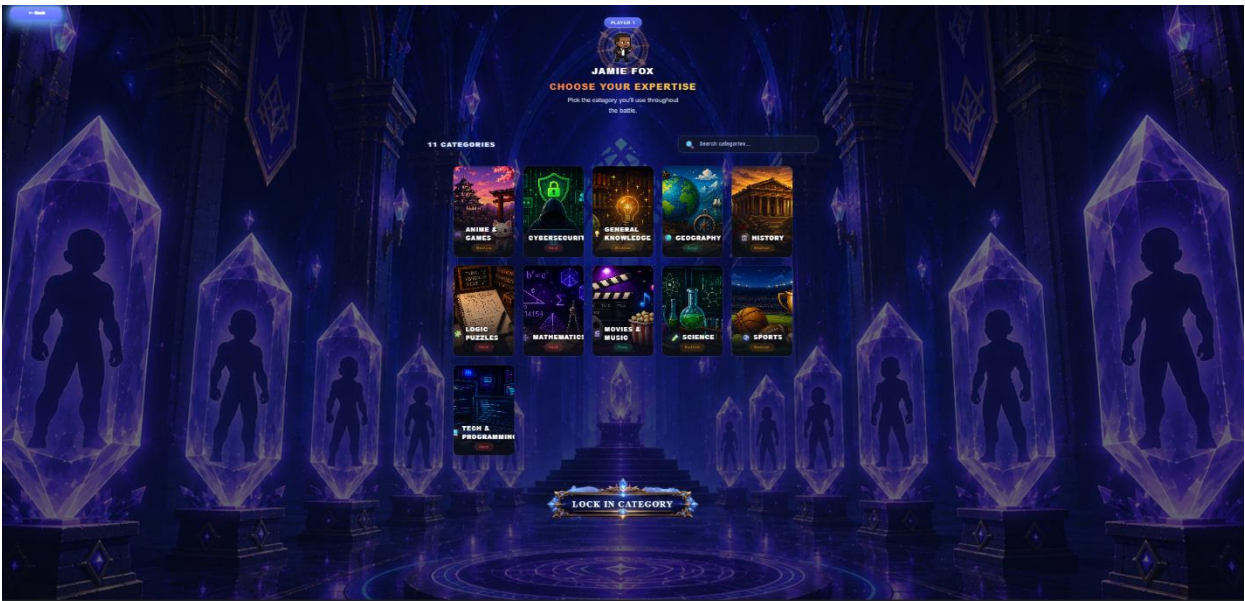


After generation completes, a summoning modal presents the newly created pixel champion, allowing players to choose a custom name and preferred voice before finalizing creation.

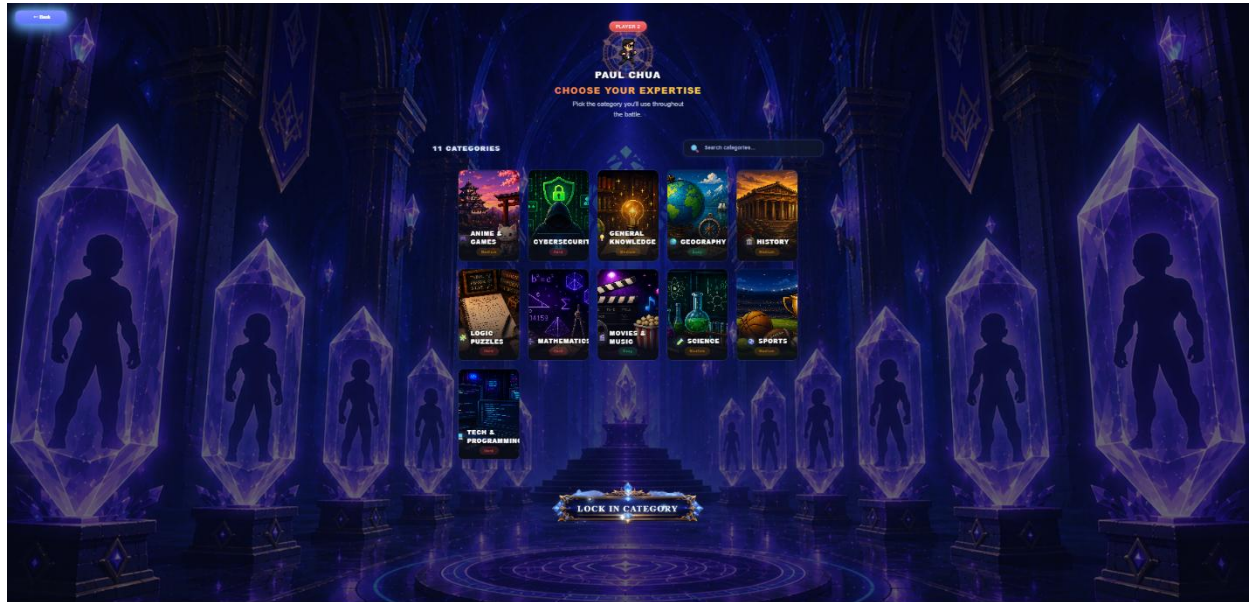


Similar to Player 1, Player 2 uploads a selfie while an animated crystal chamber displays the future fighter, combining artificial intelligence, fantasy aesthetics, and interactive customization before entering the arena.

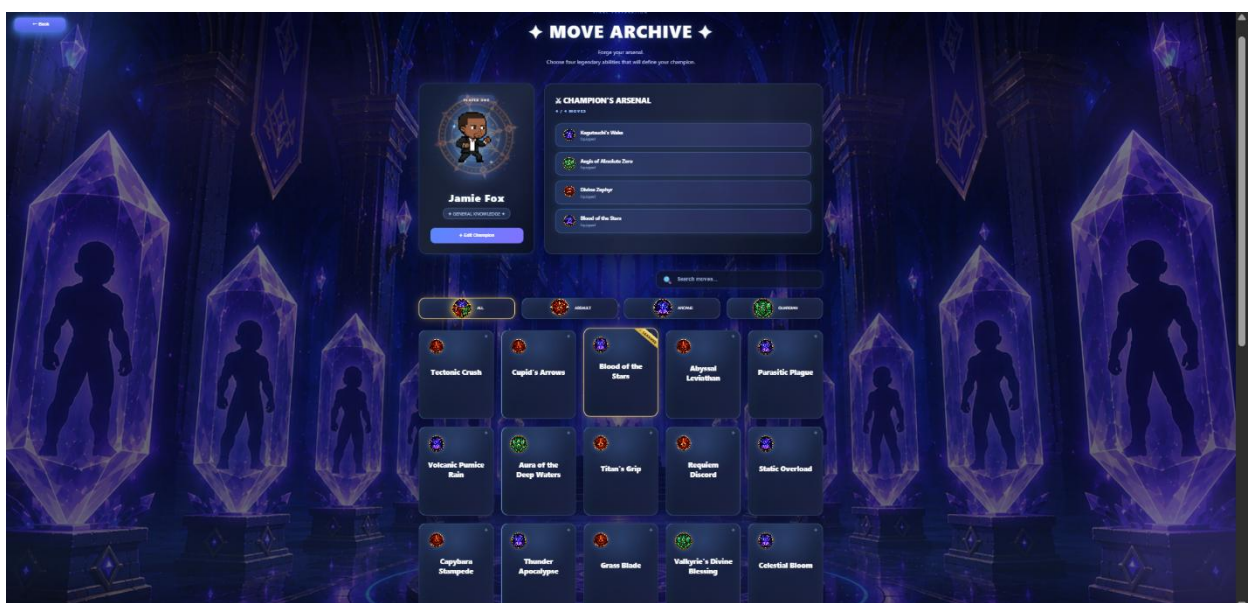
After generation completes, a summoning modal presents the newly created pixel champion, allowing players to choose a custom name and preferred voice before finalizing creation.



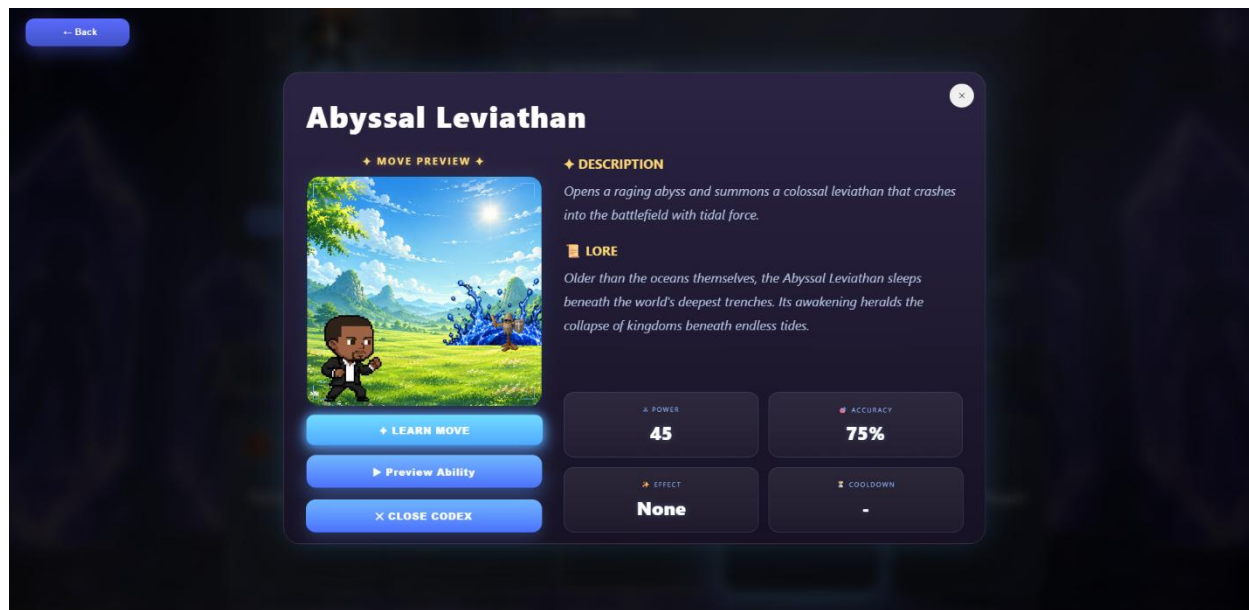
Player 1 selects a knowledge category from multiple themed cards, including cybersecurity, mathematics, history, and science, determining the expertise used throughout the upcoming battle.



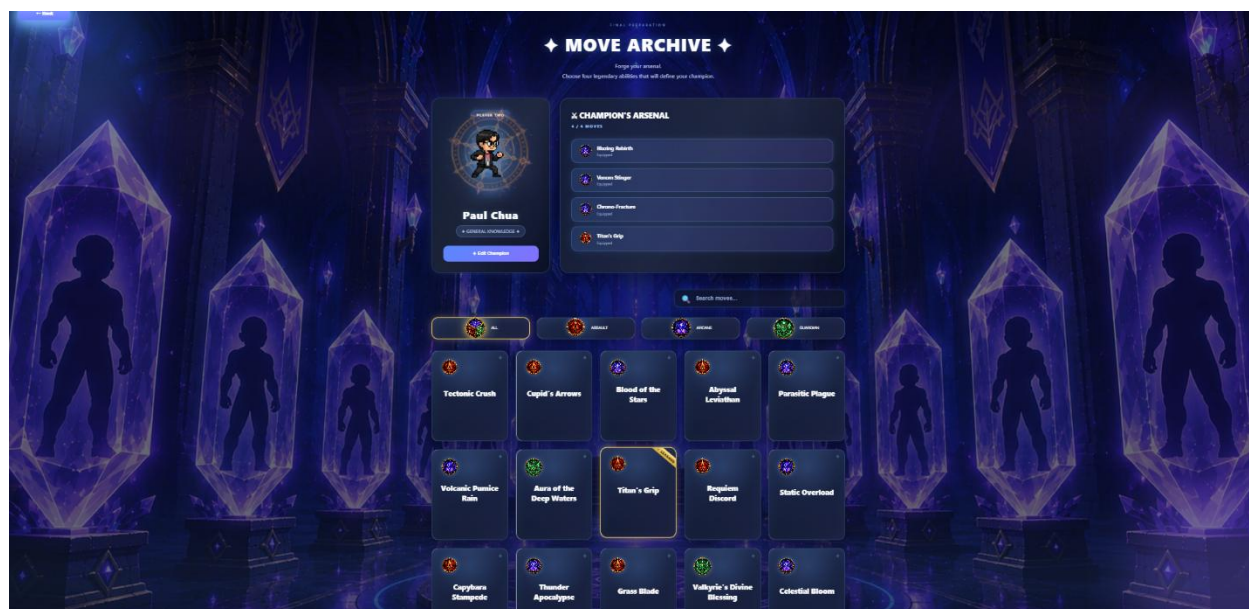
Player 2 chooses a preferred expertise category through the same magical interface, ensuring both competitors enter the arena with personalized trivia specializations and strategic advantages.



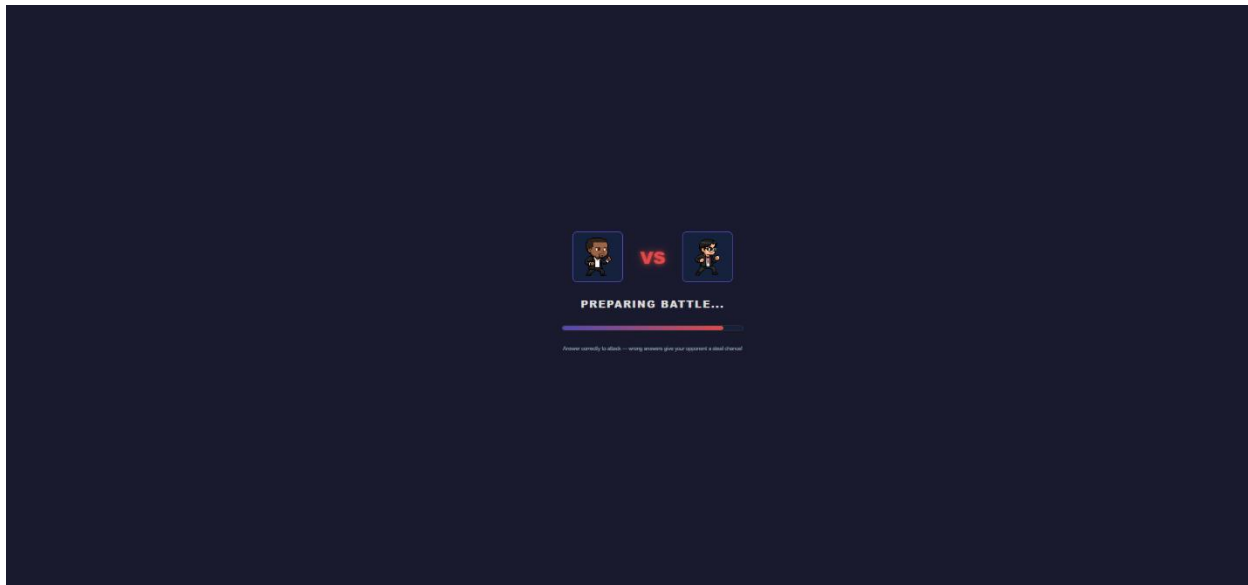
The move archive allows Player 1 to manage equipped abilities, browse legendary attacks, filter categories, and prepare a four-move arsenal before the battle begins.



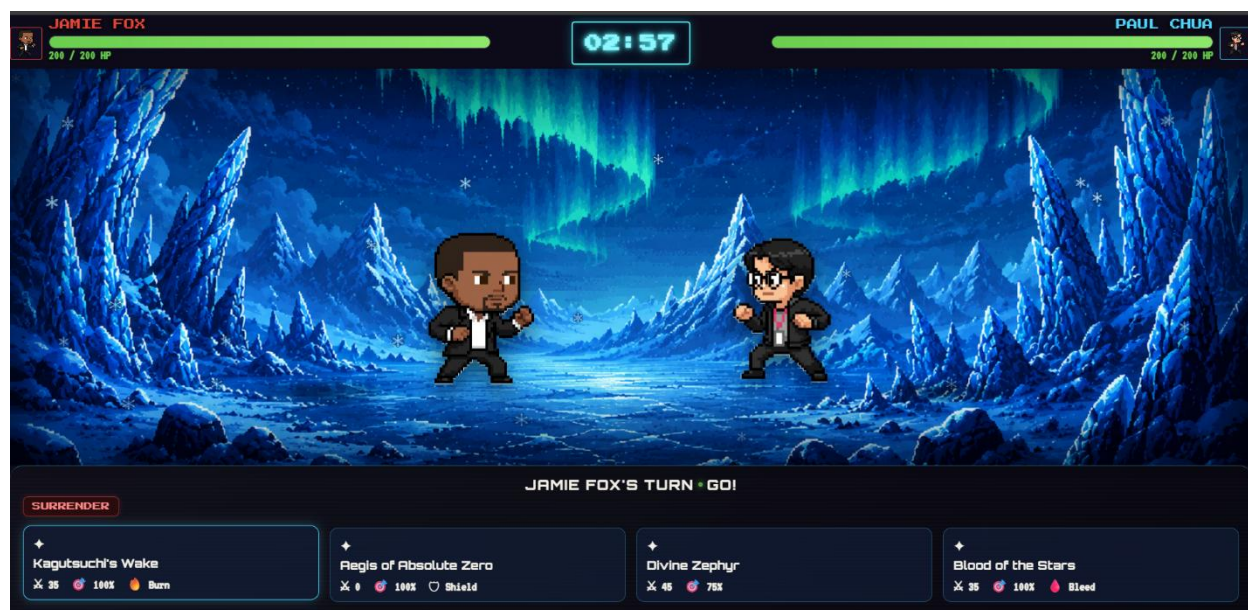
A detailed codex popup displays move artwork, lore, description, power, accuracy, effects, and cooldown information, helping players evaluate abilities before adding them permanently.



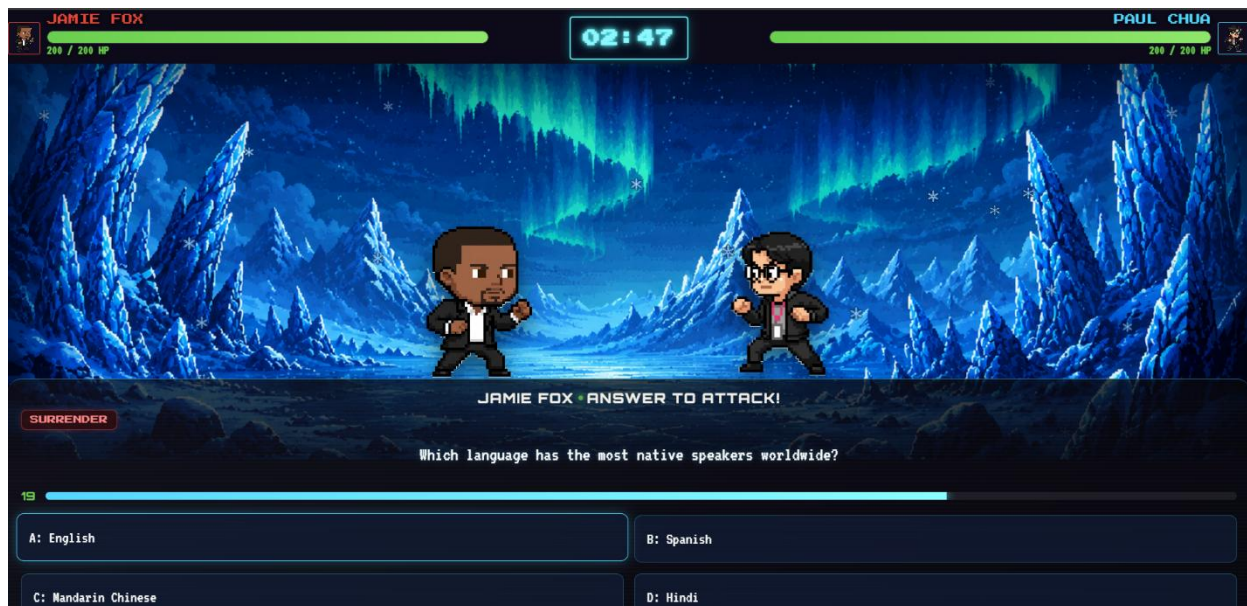
Player 2 organizes equipped abilities within the move archive, selecting offensive and supportive attacks while reviewing categories and finalizing a personalized combat loadout.



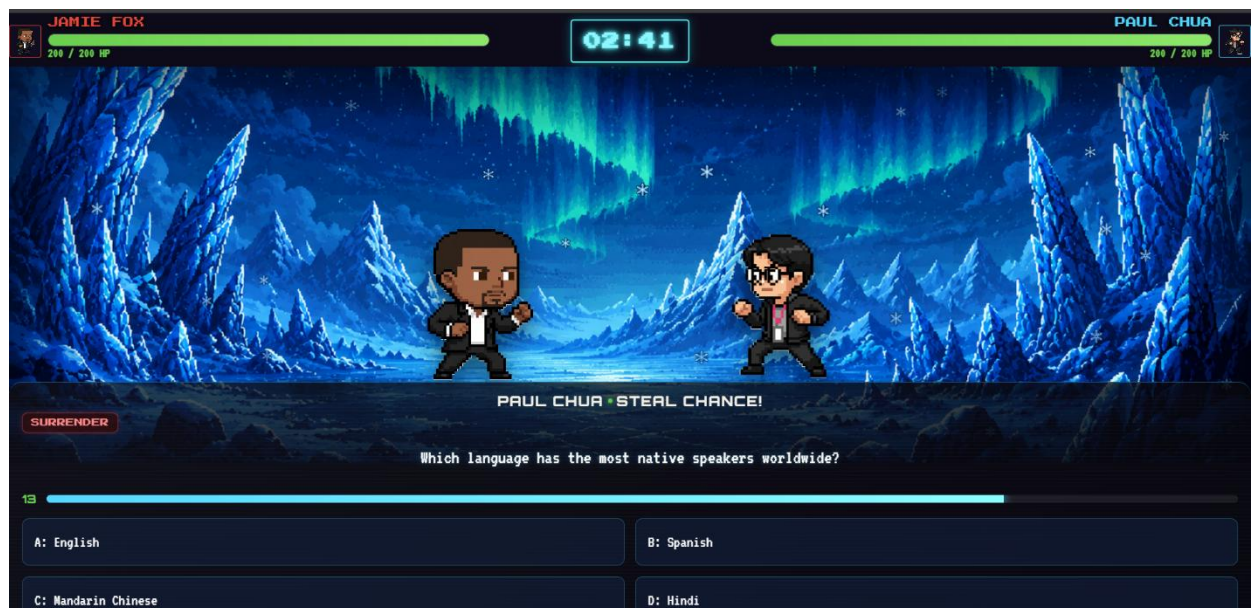
The preparation screen introduces both champions while a loading bar builds anticipation, signaling that the trivia combat sequence is about to begin.



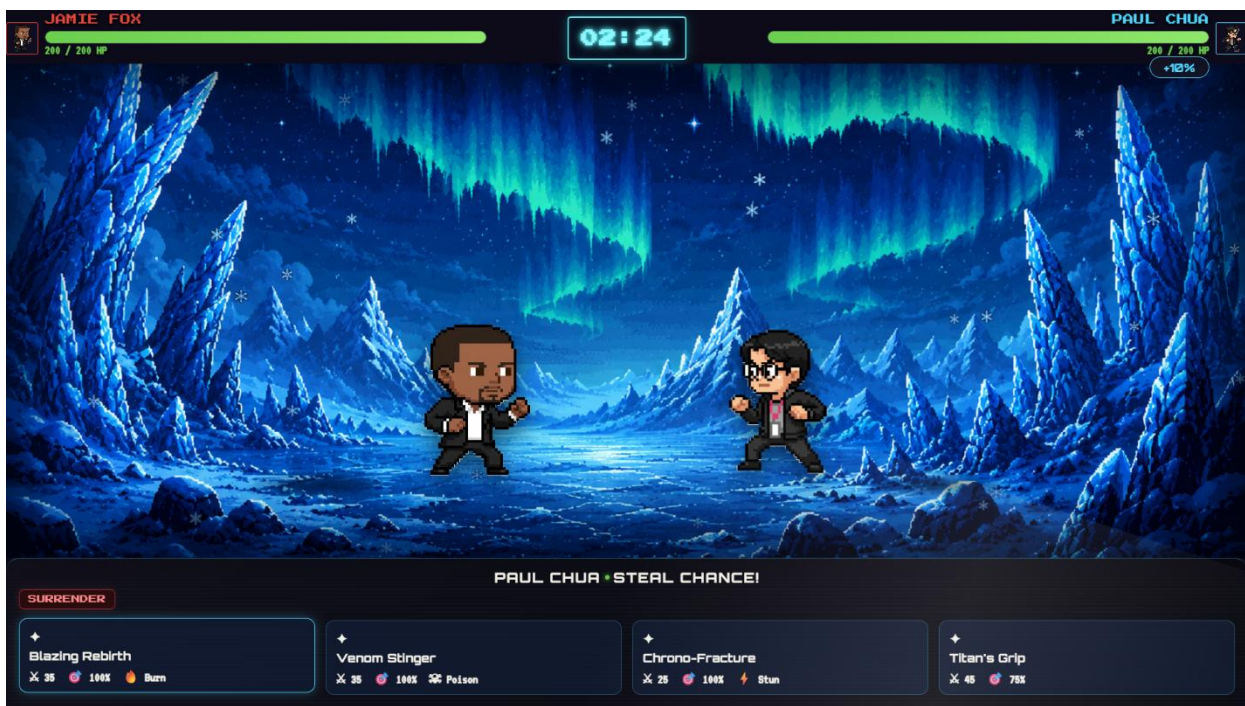
The battle arena opens with retro pixel-art visuals, animated snow, health bars, timer, and selectable moves, creating a classic fighting-game atmosphere for turn-based trivia combat.



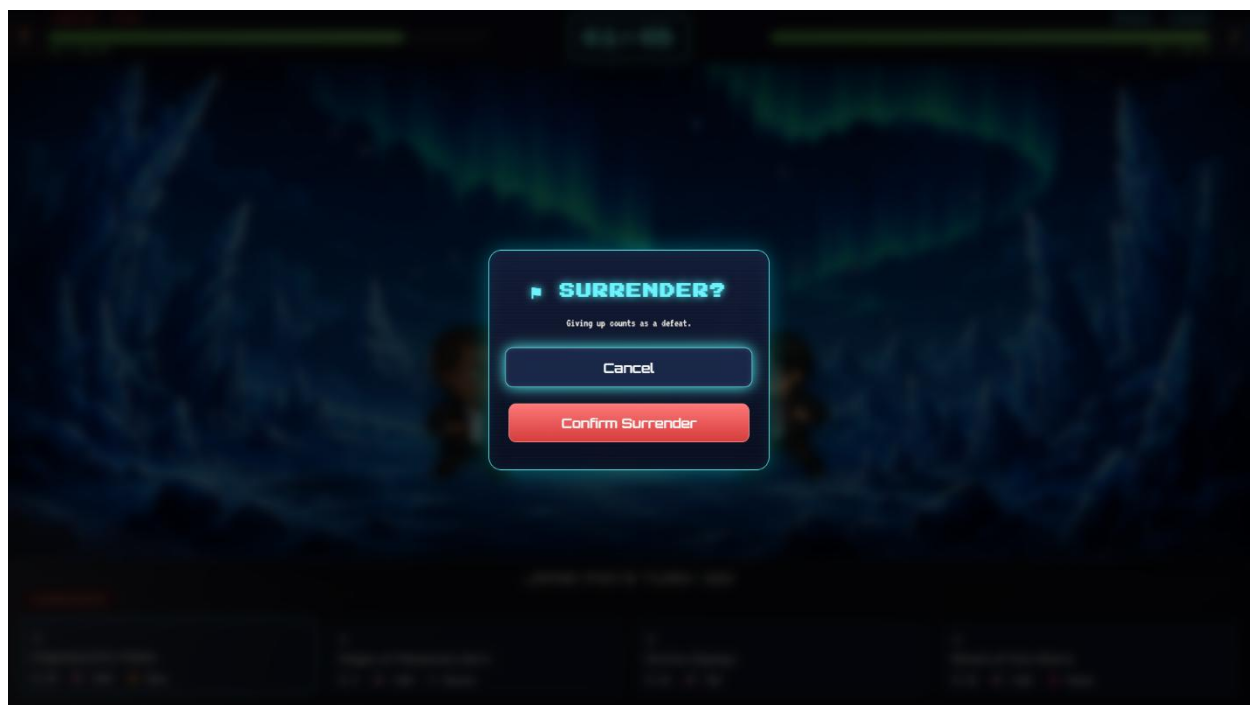
Player 1 receives a trivia question during the attack phase, with a countdown timer and multiple-choice answers determining whether the selected move successfully damages the opponent.



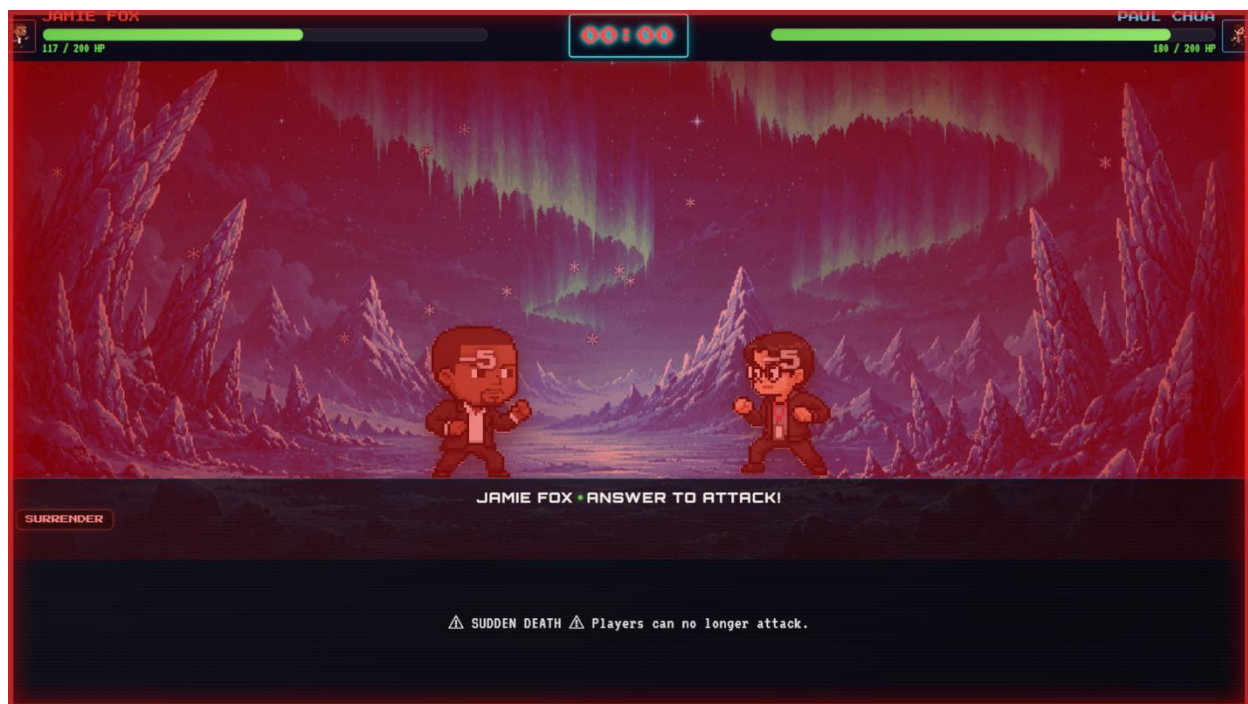
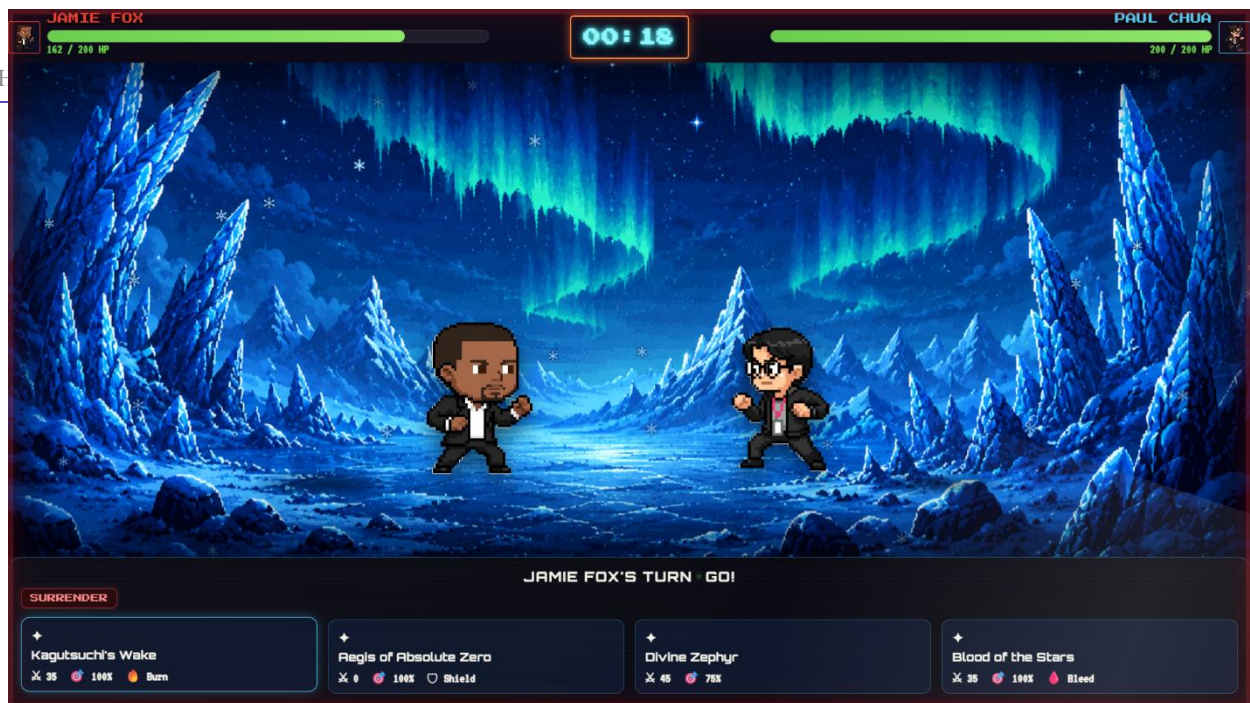
After an incorrect answer, Player 2 gains a steal opportunity, allowing the opponent to answer the same question and potentially seize momentum in the battle.

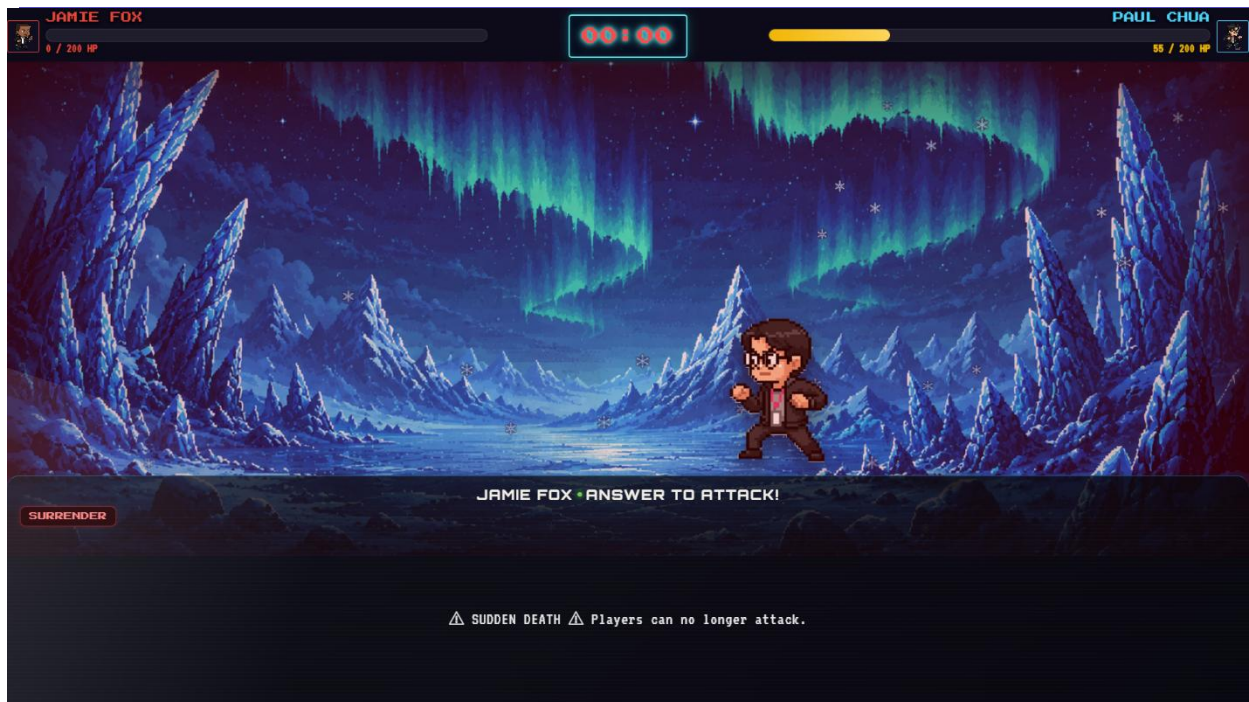


During the steal phase, Player 2 chooses a move using the recovered turn, demonstrating how failed answers can immediately create offensive opportunities for the opposing player.

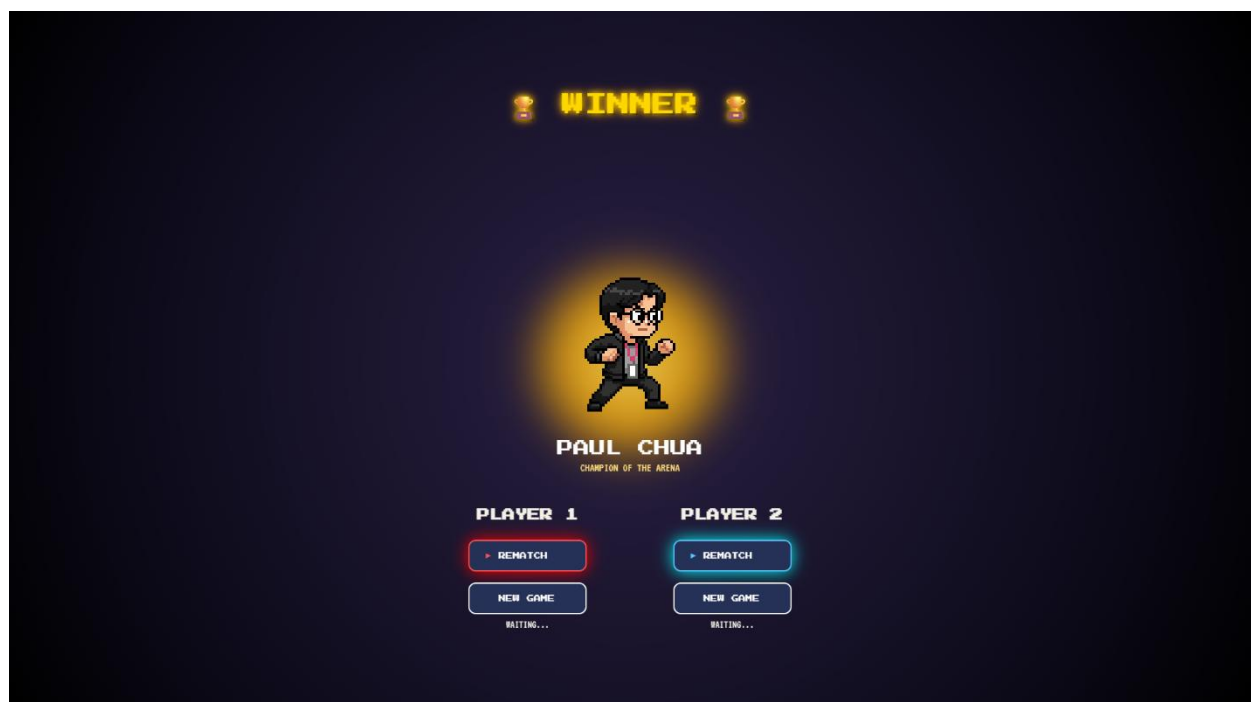


The surrender dialog pauses the match and asks for confirmation, clearly warning players that conceding the battle counts as a defeat before returning to gameplay.





The battle concludes with Player 2 standing victorious as the defeated champion disappears, presenting the decisive result after sudden death and remaining health calculations.



The winner celebration screen showcases the victorious champion with glowing effects and multiplayer rematch or new game options, providing a satisfying conclusion to the arena battle.

6. Skills & Technologies

The project is intentionally scoped to web-native technologies — no installation required to play, runs in any browser, and the skills learned are directly transferable to professional front-end web development.



Figure 5 — Four skill categories and technologies used across the project

Category	Technology	What I Learn & Build
Version Control	Git & GitHub	Branching workflow, daily commits, pull requests, GitHub releases. Tag v1.0 as final demo build.
Code Editor	VS Code + Live Server	Write and live-preview HTML/CSS/JS. Syntax highlighting, error detection, integrated terminal.
Page Structure	HTML5	Semantic elements, file input for photo uploads, multi-page navigation, div-based layout hierarchy.
Styling & Animation	CSS3	Flexbox & Grid layouts, CSS variables, @keyframes animations, transitions on HP bars, media queries for mobile, z-index layering for overlays.
Game Logic	JavaScript ES6+	Variables, functions, arrays, objects, if/else, Math.random(). DOM manipulation. Event listeners. setInterval/setTimeout for timers. JSON.parse/stringify. State machine pattern.
Data Persistence	localStorage API	Save and load player data (name, sprite, moves, streak, bonus) across multiple pages without a database or server.
Async Programming	fetch() + async/await	Call external APIs for sprite generation and question generation. Handle loading states, errors, and JSON responses gracefully.
AI Sprite Generation	Replicate API	Send a base64-encoded user photo to a pixel-art AI model. Display the returned sprite in the game. Handle API latency with a loading spinner.

Custom Questions	questions.json	Hardcoded question bank for offline/university mode. 30+ questions across 4 categories (science, geography, technology, history). Switchable via a single MODE flag.
Audio	Web Audio API	Sound effects for every game event (hit, crit, correct, wrong, steal, victory, defeat) plus looping background music. Volume control.
Testing & Debugging	Chrome DevTools	Console for JS errors, Network tab for API calls, Application tab for localStorage, Device toolbar for mobile layout testing.
Input Hardware (Optional)	Dual PlayStation2 (PS2) Controllers connected via USB adapter	Optional hardware to enable dual-controller gameplay throughout the battle system for better gaming experience

7. Project Timeline

Structured over 11 weeks at 1 hour per day (~77 hours total). Each week has a fixed deadline and a clear deliverable.

Week	Phase	Key Deliverable	Deadline
1	Foundations	Dev environment, Git repo, static HTML/CSS battle card, JS fundamentals in console	End of Day 7
2	Foundations	fetch() & async/await, localStorage, Flexbox & Grid layouts	End of Day 14
3	Character Creation	File upload UI, Replicate AI sprite pipeline, two-player localStorage save	End of Day 21
4	Move Catalogue	moves.json, catalogue grid UI, 4-move selection guard, data persists	End of Day 28
5	Question Engine	getQuestion() with custom modes, questions.json (30+), question overlay UI with timer	End of Day 35
6	Question Mechanics	Streak counter (crit at 3), steal phase (+10% bonus), full question flow end-to-end	End of Day 42
7	Battle Engine	calculateDamage(), turn state machine, full battle screen UI, first playable build	End of Day 49
8	Effects & Sound	CSS animation library, 9+ sound effects, HP colour shifts, question feedback visuals	End of Day 56
9	Polish	Entrance cinematics, win/lose screen, 3-min timer + sudden death drain, balance tuning	End of Day 63
10	Testing & Deploy	Think-aloud user testing (3 testers), Vercel deployment, HCI report draft	End of Day 70
11	Final Submission	Bug sweep, v1.0 GitHub release, complete HCI report, 5-min demo rehearsal	End of Day 77

8. Risk Assessment & Contingency Plan

Risk	Likelihood	Mitigation
Replicate API costs money or hits rate limits	Medium	Use a placeholder PNG sprite during development. Only call the real API a few times to test. If cost is prohibitive, pre-generate sprites and cache them locally.
3-minute timer fires mid-question	Low	Check a suddenDeathActive flag at turn start. If true, finish the current question first, then trigger sudden death immediately after — never interrupt mid-answer.
Streak/steal state desync	Medium	All game state lives in one battleState object in battle.js. Never update streak or bonus in the UI layer. Test all edge cases explicitly in Week 8.
Two-player handoff confuses testers	Medium	Full-screen overlay between every turn. Conduct think-aloud user testing in Week 10 specifically watching for handoff confusion points.
Running out of time in Week 11	Low	Entrance cinematic (30 min) is the first thing to cut. Core mechanics (question, steal, streak, timer) are scoped to Weeks 5–6 and 9 with buffer built in.

9. Conclusion

Trivia Arena is a practical, buildable, and academically substantive HCI project for audience of any ages. It addresses a genuine engagement problem in quiz-based learning tools using established game design and HCI principles. Every mechanic — from the 25-second question timer to the steal phase to the sudden death drain — is a conscious interaction design decision that can be justified, evaluated through user testing, and written about in a report.

The project is technically achievable for a beginner developer within 11 weeks at 1 hour per day. It uses only web-native technologies, produces a working, playable game that can be demonstrated live to any audience.

Usability Goals	Achieve this goal. (Y / N/ NA)	User Experience Goals	Achieve this goal. (Y / N/ NA)	Design Principles	Achieve this principle. (Y / N/ NA)
Effectiveness	Y	Satisfying	Y	Visibility	Y
Efficiency	Y	Enjoyable	Y	Feedback	Y
Safety	Y	Helpful	Y	Constraints	Y
Utility	Y	Motivating	Y	Consistency	Y
Learnability	Y	Aesthetically pleasing	Y	Mapping	Y
Memorability	Y	Supportive of creativity	Y	Affordance	Y
		Emotionally fulfilling	Y		